Homework 6

EE 999 — Introduction to Control Systems

Patrick Gordon • October 26, 2026

Due Wednesday, October 28, 2026

Problem 1.A — Text problem 3.19(a)

Find the transfer function for the block diagram in Fig. 3.50(a).

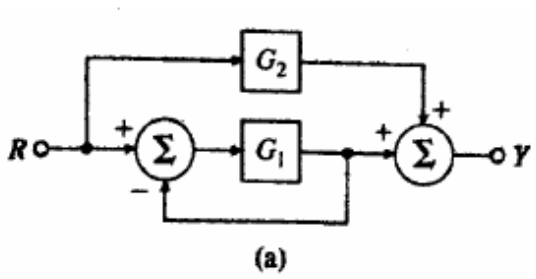


Figure 1: Text figure 3.50(a) (8th Ed.).

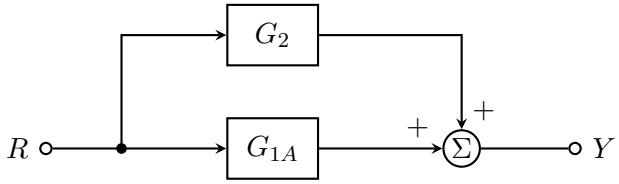


Figure 2: Fig. 3.50(a) with the inner summing junction and G_1 combined into G_{1A}.

$$G_{1A} = \frac{G_1}{1 + G_1} \tag{1}$$

$$\frac{Y}{R} = G_2 + G_{1A} \tag{2}$$

$$= G_2 + \frac{G_1}{1 + G_1} \tag{3}$$

Problem 1.B — Text problem 3.19(b)

Find the transfer function for the block diagram in Fig. 3.50(b).

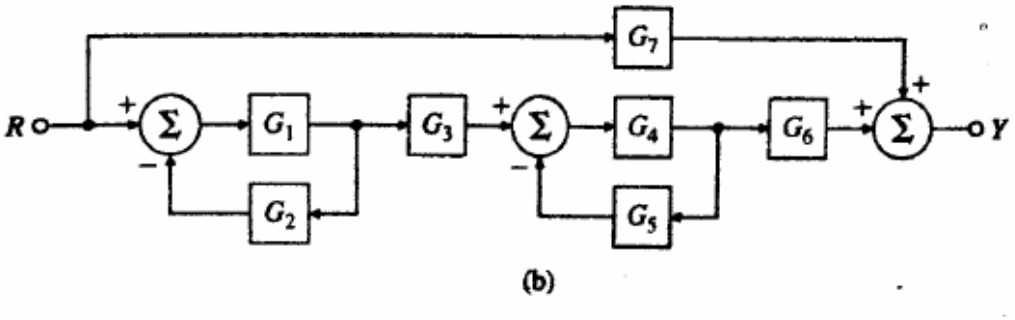


Figure 3: Text figure 3.50(b) (8th Ed.).

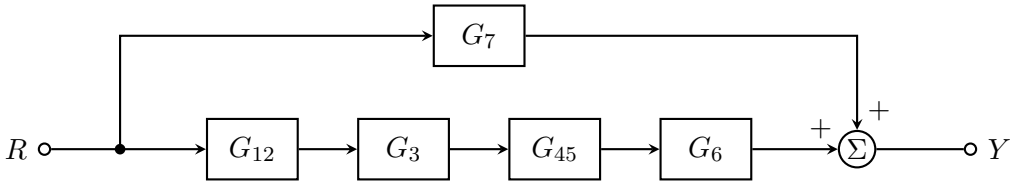


Figure 4: Fig. 3.50(b) with the first summing junction, G_1 and G_2 combined into G_{12}, and the second summing junction, G_4 and G_5 combined into G_{45}.

$$G_{12} = \frac{G_1}{1 + G_1 G_2} \tag{4}$$

$$G_{45} = \frac{G_4}{1 + G_4 G_5} \tag{5}$$

$$\frac{Y}{R} = G_7 + G_{12} G_3 G_{45} G_6 \tag{6}$$

$$= G_7 + \frac{G_1 G_3 G_4 G_6}{(1 + G_1 G_2)(1 + G_4 G_5)} \tag{7}$$